Stat 380 Final

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Predicting Drowsy Driving: Can we Machine Learning Model to be able to Predict Drowsiness in Drivers?

Introduction

Intuition Behind the Choice of Data Set

- The data set must be large to be able to get a good testing pool.
- The data set has meaningful predictors are practically relevant to our test case.
- The data set contains a drowsy/awake variable
- The data set follows **F.A.I.R.** and **C.A.R.E.** principles.

Ethical Considerations in the Choice of our Data Set

In order to ensure the ethical feasibility of the data set, we also scrutinized our selection to follow the **F.A.I.R.** and **C.A.R.E.** principles as discussed in class.

Since we had obtained our data from the open-source data hub Kaggle, we can ensure that anyone can **find** and **access** this data set through Kaggle's use guidelines, stating for a free distribution of data sets that are published for public use to practice ML or data analysis on. The data is also highly versatile, taking a csv format. This is one of the most **interoperable** formats there are for data sets, especially for our use-case in R, where reading in a csv is already built into the base. Lastly, the data set can be **reused** for multiple types of data analysis, and will always remain relevant for its time period.

In terms of **collective benefit**, the data set contributes to an overall understanding of driving without harming any individuals , due to the fact that it is simply raw data. Secondly, as I had stated the author's name, there is clearly ownership over the data set coming from kaggle as mentioned, showing proper **Authority to Control**. Since proper attributions are maintained, it also falls under the **Responsibility and Ethics** guidelines of the C.A.R.E. principles.

The dataset can be found here.

Cleaning

Load initial libraries

Note that the dataset name is Drowsy_datset

Table 1: First 5 NATURAL Train Files:

File
Drowsy_datset/train/NATURAL/1000.png
Drowsy_datset/train/NATURAL/1001.png
Drowsy_datset/train/NATURAL/1002.png
Drowsy_datset/train/NATURAL/1003.jpg_0.png
Drowsy_datset/train/NATURAL/1004.png

Table 2: First 5 DROWSY Train Files:

File
Drowsy_datset/train/DROWSY/1.jpg_0.png
Drowsy_datset/train/DROWSY/1.png
Drowsy_datset/train/DROWSY/10.jpg_0.png
Drowsy_datset/train/DROWSY/10.png
Drowsy_datset/train/DROWSY/100.png

Table 3: First 5 NATURAL Test Files:

File
Drowsy_datset/test/NATURAL/v1013.png
Drowsy_datset/test/NATURAL/w0002.png
Drowsy_datset/test/NATURAL/w0003.png
Drowsy_datset/test/NATURAL/w0004.png
Drowsy_datset/test/NATURAL/w0005.png

Table 4: First 5 DROWSY Test Files:

File
Drowsy_datset/test/DROWSY/1.png
Drowsy_datset/test/DROWSY/10.png
Drowsy_datset/test/DROWSY/100.jpg_0.png

File
Drowsy_datset/test/DROWSY/102.jpg_0.png
Drowsy_datset/test/DROWSY/105.jpg_0.png

This is showing that some files are PNG and some are JPEG.

```
Drowsy_datset/train/DROWSY/1.jpg_0.png  => PNG check:  TRUE
Drowsy_datset/train/DROWSY/1.png  => PNG check:  FALSE
Drowsy_datset/train/DROWSY/10.jpg_0.png  => PNG check:  TRUE
Drowsy_datset/train/DROWSY/10.png  => PNG check:  FALSE
Drowsy_datset/train/DROWSY/100.png  => PNG check:  FALSE
```

```
Drowsy_datset/train/DROWSY/1.jpg_0.png  => JPEG check:  FALSE
Drowsy_datset/train/DROWSY/1.png  => JPEG check:  TRUE
Drowsy_datset/train/DROWSY/10.jpg_0.png  => JPEG check:  FALSE
Drowsy_datset/train/DROWSY/10.png  => JPEG check:  TRUE
Drowsy_datset/train/DROWSY/100.png  => JPEG check:  TRUE
```

We'll have handle BOTH. The following code chunk loads ALL of the images into a drowsy and natural set, by handling png or jpeg (our data cleaning)

We combine the datasets into one train and test dataset.

We look at the initial class imbalance. Notice its not that bad.

```
    label    n
1  Drowsy 2809
2 Natural 3050
```

```
    label   n
1  Drowsy 757
2 Natural 726
```

We shuffle the dataset

Preprocessing

We need to ensure that the datasets are the same going in. This applies CLAHE and HFE to apply pre-processing.

The first augmentation technique we use therefore is **Contrast Limited Adaptive Histogram Equalization (CLAHE)**. This is a refined version of normal Histogram Equalization that contains a regularization parameter to mitigate the strength of the initial histogram equalization technique by adding a clipping feature to the initial dataset, ensuring that contrasts are never set too high. Furthermore, the image is subdivided into separate sections, where histogram equalization is applied separately. These sections are then combined using bilinear interpolation, which ensures a smooth transition between each segmented histogram. This way, we do not create entire areas of extreme contrast and intensity, something that normal histogram equalization is prone to doing. We therefore opted to maintain this augmentation technique in our pipeline.

Lastly, **High-Frequency Emphasis (HFE)** is a type of processing technique that boosts subtle details and fine textured by enhancing the frequency components of a given image. The "frequency" is usually represented via a Fourier Transform into many different frequency elements, where low frequencies correlated to slowly varying information (think features that are fairly consistent throughout an image) and high frequencies relating to contrasting sections of an image (areas where features abruptly change). This allows for the convolutions in computer vision to better pick up patterns and improve generalizability. After carefully analyzing our own parameters and the differences between our and our parents' methodologies, we opted to maintain this data processing technique, as it is vital for the extrapolation of the plant diseases. For our high frequency emphasis, we use the following Gaussian blur for convolutions with a sliding of 1. (THIS IS NOT DEEP LEARNING, SIMPLY A TRANSFORMATION WITH STATIC CONVOLVING FILTERS, NOT LEARNABLE FILTERS).

$$\frac{1}{10}\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

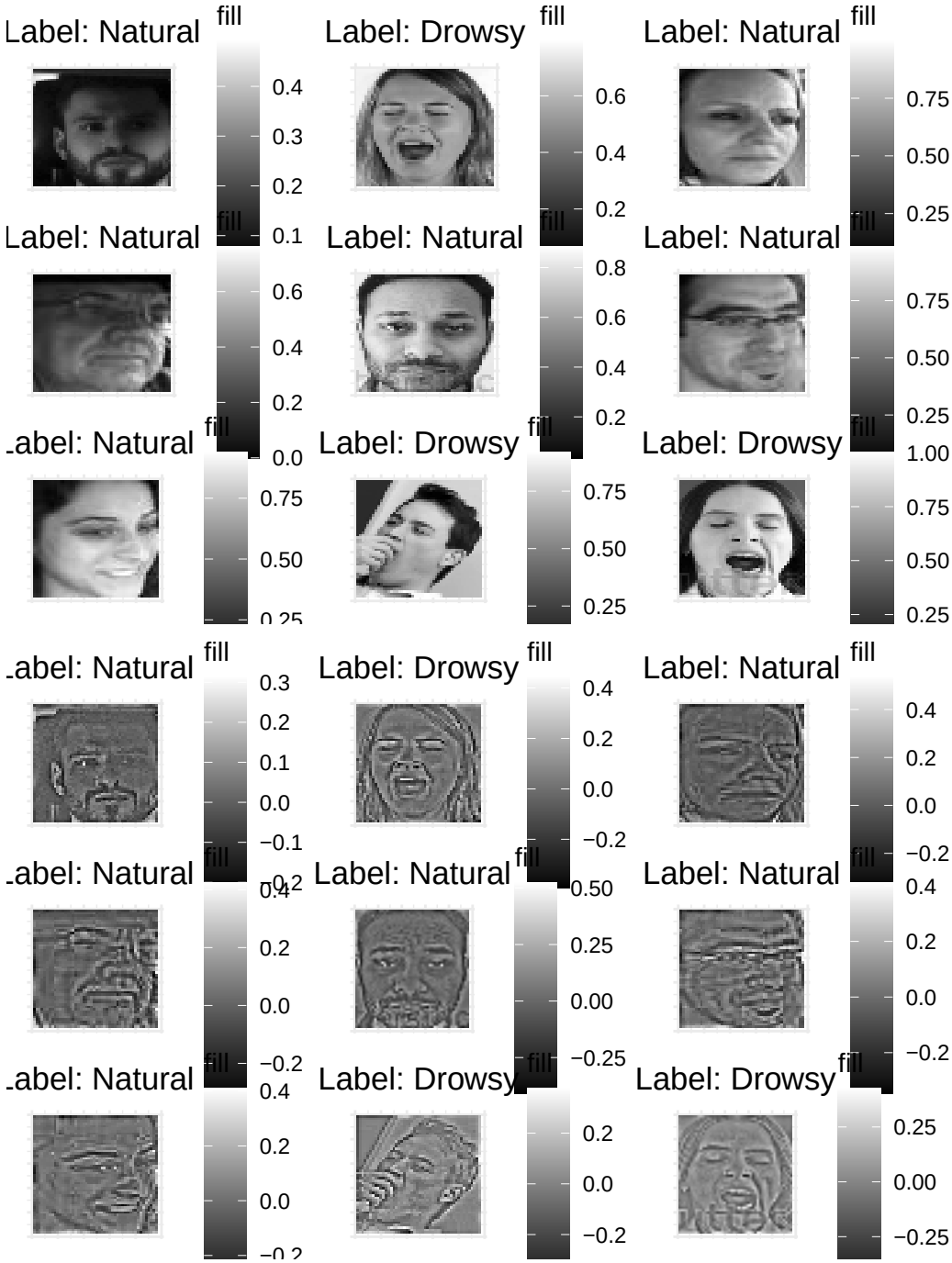
Readying for models

We define our new models as one with feature engineering and one without feature engineering. We stretch and flatten our images into one long string of vectors.

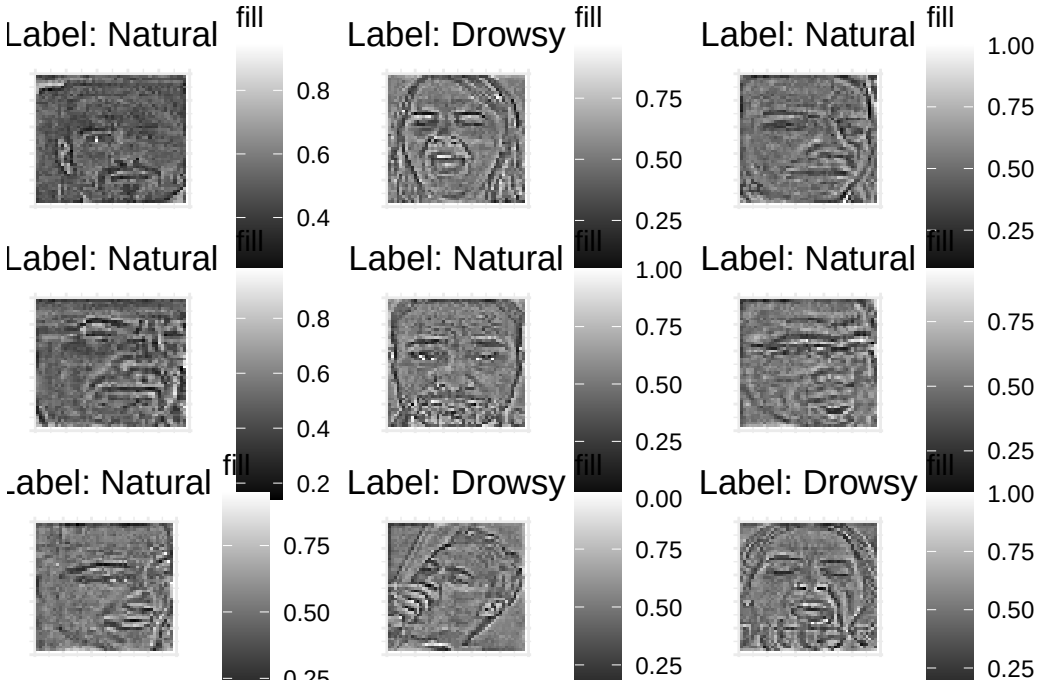
These are plotting functions. This one reconstructs the image from the flattened array.

This one plots the image.

We print the first 9. Notice the differences.



We need to change the scaling back to [0,1] after feature engineering.



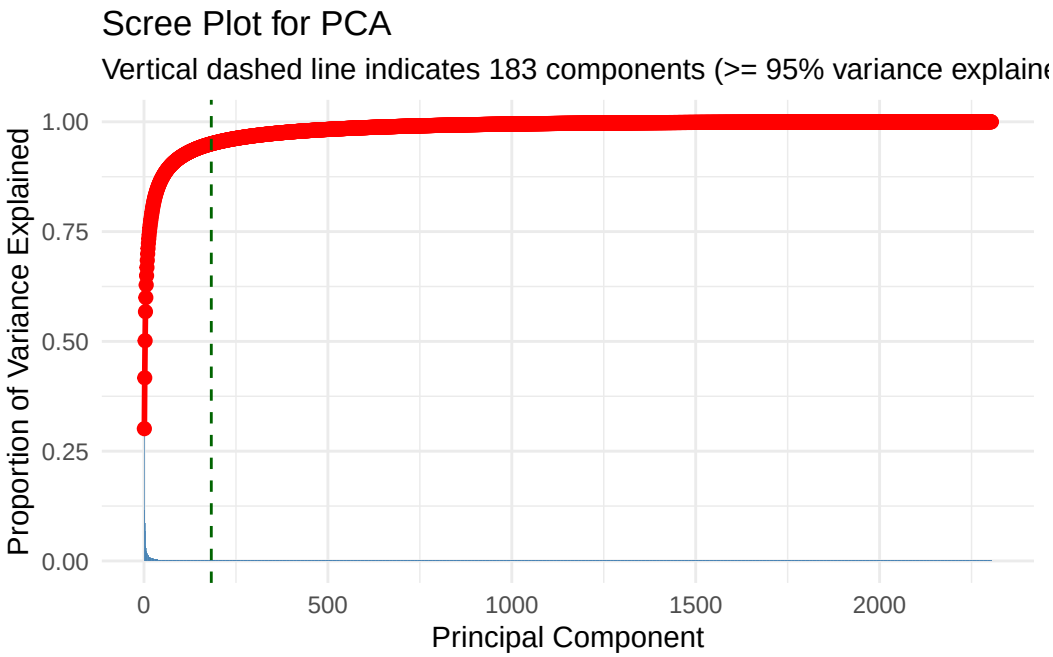
Save our datasets here.

Do PCA

we will do PCA. We will use the PCA done on the train set to transform our test set to then evaluate with our models later.

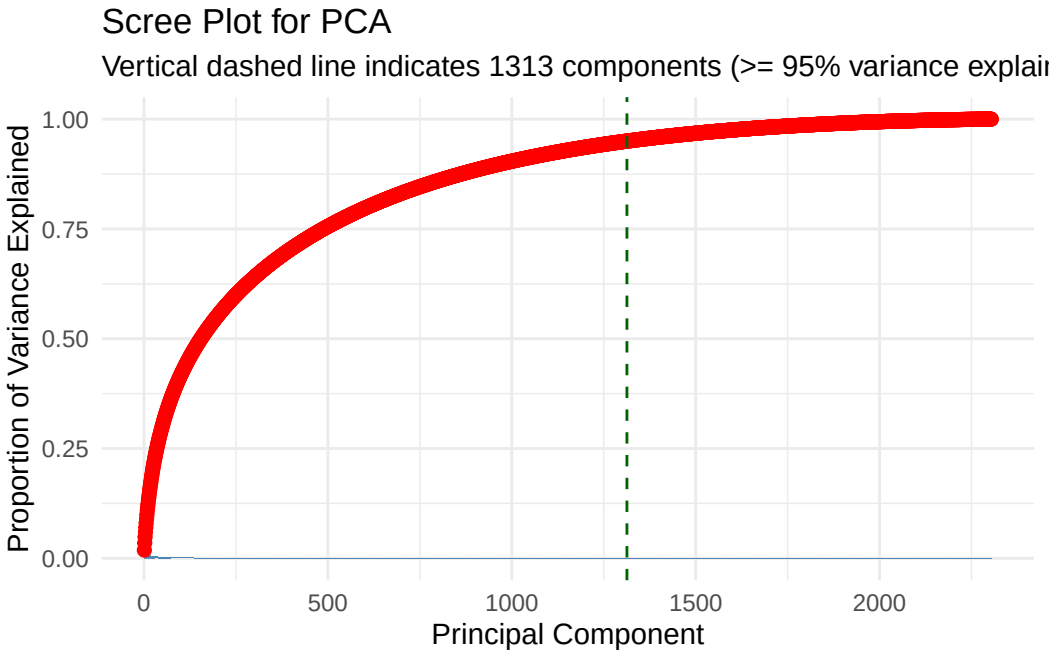
For no pre-processing

```
Number of components to reach at least 95% variance explained: 183
```



For pre-processed, we also perform pca.

```
Number of components to reach at least 95% variance explained: 1313
```



The contrast actually increases the variance a great deal. While this might be helpful for complex models, for simple ML models it could hurt it.

Model Ablation Study

I didn't take out the libraries. Its more important to run the models right now.All of these have their own seperate hyperparameter search and are 5 fold cv'd. We save a confusion matrix, some metrics like accuracy, recall, precision, f1 score, AUC from ROC, the actual ROC data points, and a png of the ggplot of the AUC ROC per model. The actual ROC data points can be used to display multiple model AUCs on one plot, if we so choose to later. Lastly, we save the models as .rdl. We will need this to open them back up for our stacking model.

Gaussian SVM, No feature pre-processing, PCA

```
        sigma    C  Accuracy     Kappa  AccuracySD    KappaSD
1 0.003223246 0.25 0.9168799 0.8334783 0.015271047 0.03062410
2 0.003223246 0.50 0.9385558 0.8768799 0.009290708 0.01865789
3 0.003223246 1.00 0.9569890 0.9138174 0.008287259 0.01662576

        sigma C
3 0.003223246 1

Model saved as 'Base_Models/SVMModelNoFeatureEngineeringPCA.rds'. To load it later, use:

loaded_model <- readRDS('Base_Models/SVMModelNoFeatureEngineeringPCA.rds')

BestTune hyperparameters saved as 'Base_Models_Data/SVM/BestHyperparams_SVMNoFeatureEngineeri

Confusion Matrix and Statistics

          Reference
Prediction Drowsy Natural
   Drowsy     278       0
   Natural    117     346

               Accuracy : 0.8421
                 95% CI : (0.8138, 0.8676)
    No Information Rate : 0.5331
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.6893
```

```
 Mcnemar's Test P-Value : < 2.2e-16

            Sensitivity : 0.7038
            Specificity : 1.0000
         Pos Pred Value : 1.0000
         Neg Pred Value : 0.7473
             Prevalence : 0.5331
         Detection Rate : 0.3752
   Detection Prevalence : 0.3752
      Balanced Accuracy : 0.8519

       'Positive' Class : Drowsy


Confusion matrix saved as 'Base_Models_Data/SVM/ConfusionMatrix_SVMNoFeatureEngineeringPCA.cs

Performance metrics saved as 'Base_Models_Data/SVM/Metrics_SVMNoFeatureEngineeringPCA.csv'.
ROC Data for plot building saved as 'Base_Models_Data/SVM/ROCData_SVMNoFeatureEngineeringPCA

ROC plot saved as 'Base_Models_Data/SVM/ROC_SVMNoFeatureEngineeringPCA.png'.
```

XGBoost, No feature pre-processing, PCA

	eta	max_depth	gamma	colsample_bytree	min_child_weight	subsample	nrounds
1	0.3	1	0	0.6	1	0.50	50
4	0.3	1	0	0.6	1	0.75	50
7	0.3	1	0	0.6	1	1.00	50
10	0.3	1	0	0.8	1	0.50	50
13	0.3	1	0	0.8	1	0.75	50
16	0.3	1	0	0.8	1	1.00	50
55	0.4	1	0	0.6	1	0.50	50
58	0.4	1	0	0.6	1	0.75	50
61	0.4	1	0	0.6	1	1.00	50
64	0.4	1	0	0.8	1	0.50	50
67	0.4	1	0	0.8	1	0.75	50
70	0.4	1	0	0.8	1	1.00	50
19	0.3	2	0	0.6	1	0.50	50
22	0.3	2	0	0.6	1	0.75	50
25	0.3	2	0	0.6	1	1.00	50
28	0.3	2	0	0.8	1	0.50	50

31	0.3	2	0	0.8	1	0.75	50
34	0.3	2	0	0.8	1	1.00	50
73	0.4	2	0	0.6	1	0.50	50
76	0.4	2	0	0.6	1	0.75	50
79	0.4	2	0	0.6	1	1.00	50
82	0.4	2	0	0.8	1	0.50	50
85	0.4	2	0	0.8	1	0.75	50
88	0.4	2	0	0.8	1	1.00	50
37	0.3	3	0	0.6	1	0.50	50
40	0.3	3	0	0.6	1	0.75	50
43	0.3	3	0	0.6	1	1.00	50
46	0.3	3	0	0.8	1	0.50	50
49	0.3	3	0	0.8	1	0.75	50
52	0.3	3	0	0.8	1	1.00	50
91	0.4	3	0	0.6	1	0.50	50
94	0.4	3	0	0.6	1	0.75	50
97	0.4	3	0	0.6	1	1.00	50
100	0.4	3	0	0.8	1	0.50	50
103	0.4	3	0	0.8	1	0.75	50
106	0.4	3	0	0.8	1	1.00	50
2	0.3	1	0	0.6	1	0.50	100
5	0.3	1	0	0.6	1	0.75	100
8	0.3	1	0	0.6	1	1.00	100
11	0.3	1	0	0.8	1	0.50	100
14	0.3	1	0	0.8	1	0.75	100
17	0.3	1	0	0.8	1	1.00	100
56	0.4	1	0	0.6	1	0.50	100
59	0.4	1	0	0.6	1	0.75	100
62	0.4	1	0	0.6	1	1.00	100
65	0.4	1	0	0.8	1	0.50	100
68	0.4	1	0	0.8	1	0.75	100
71	0.4	1	0	0.8	1	1.00	100
20	0.3	2	0	0.6	1	0.50	100
23	0.3	2	0	0.6	1	0.75	100
26	0.3	2	0	0.6	1	1.00	100
29	0.3	2	0	0.8	1	0.50	100
32	0.3	2	0	0.8	1	0.75	100
35	0.3	2	0	0.8	1	1.00	100
74	0.4	2	0	0.6	1	0.50	100
77	0.4	2	0	0.6	1	0.75	100
80	0.4	2	0	0.6	1	1.00	100
83	0.4	2	0	0.8	1	0.50	100
86	0.4	2	0	0.8	1	0.75	100

89	0.4	2	0	0.8	1	1.00	100
38	0.3	3	0	0.6	1	0.50	100
41	0.3	3	0	0.6	1	0.75	100
44	0.3	3	0	0.6	1	1.00	100
47	0.3	3	0	0.8	1	0.50	100
50	0.3	3	0	0.8	1	0.75	100
53	0.3	3	0	0.8	1	1.00	100
92	0.4	3	0	0.6	1	0.50	100
95	0.4	3	0	0.6	1	0.75	100
98	0.4	3	0	0.6	1	1.00	100
101	0.4	3	0	0.8	1	0.50	100
104	0.4	3	0	0.8	1	0.75	100
107	0.4	3	0	0.8	1	1.00	100
3	0.3	1	0	0.6	1	0.50	150
6	0.3	1	0	0.6	1	0.75	150
9	0.3	1	0	0.6	1	1.00	150
12	0.3	1	0	0.8	1	0.50	150
15	0.3	1	0	0.8	1	0.75	150
18	0.3	1	0	0.8	1	1.00	150
57	0.4	1	0	0.6	1	0.50	150
60	0.4	1	0	0.6	1	0.75	150
63	0.4	1	0	0.6	1	1.00	150
66	0.4	1	0	0.8	1	0.50	150
69	0.4	1	0	0.8	1	0.75	150
72	0.4	1	0	0.8	1	1.00	150
21	0.3	2	0	0.6	1	0.50	150
24	0.3	2	0	0.6	1	0.75	150
27	0.3	2	0	0.6	1	1.00	150
30	0.3	2	0	0.8	1	0.50	150
33	0.3	2	0	0.8	1	0.75	150
36	0.3	2	0	0.8	1	1.00	150
75	0.4	2	0	0.6	1	0.50	150
78	0.4	2	0	0.6	1	0.75	150
81	0.4	2	0	0.6	1	1.00	150
84	0.4	2	0	0.8	1	0.50	150
87	0.4	2	0	0.8	1	0.75	150
90	0.4	2	0	0.8	1	1.00	150
39	0.3	3	0	0.6	1	0.50	150
42	0.3	3	0	0.6	1	0.75	150
45	0.3	3	0	0.6	1	1.00	150
48	0.3	3	0	0.8	1	0.50	150
51	0.3	3	0	0.8	1	0.75	150
54	0.3	3	0	0.8	1	1.00	150

93	0.4	3	0	0.6	1	0.50	150
96	0.4	3	0	0.6	1	0.75	150
99	0.4	3	0	0.6	1	1.00	150
102	0.4	3	0	0.8	1	0.50	150
105	0.4	3	0	0.8	1	0.75	150
108	0.4	3	0	0.8	1	1.00	150

	Accuracy	Kappa	AccuracySD	KappaSD
1	0.8341005	0.6678576	0.006706727	0.013568831
4	0.8334178	0.6663916	0.008433888	0.016975558
7	0.8335885	0.6668810	0.009425174	0.018878108
10	0.8346147	0.6690788	0.007866962	0.016149212
13	0.8370023	0.6736488	0.009797013	0.019854400
16	0.8352945	0.6703281	0.010858803	0.021970478
55	0.8513390	0.7022932	0.004279034	0.008371434
58	0.8528760	0.7053969	0.014130758	0.028495919
61	0.8457067	0.6911156	0.008255666	0.016740621
64	0.8520223	0.7036921	0.009631591	0.019568392
67	0.8462194	0.6921755	0.014040092	0.028082575
70	0.8460483	0.6917930	0.013466310	0.027194067
19	0.9035673	0.8069269	0.007262792	0.014415217
22	0.9054460	0.8106465	0.005448663	0.010898733
25	0.9016903	0.8032223	0.004555286	0.009172666
28	0.9013496	0.8025341	0.005781229	0.011514920
31	0.9033959	0.8065586	0.005522517	0.011011999
34	0.9033962	0.8066276	0.002139974	0.004305186
73	0.9076640	0.8151350	0.004267497	0.008598084
76	0.9132956	0.8263928	0.004079812	0.008222650
79	0.9161962	0.8322481	0.007217576	0.014568601
82	0.9037384	0.8073935	0.009933033	0.019866616
85	0.9093709	0.8185294	0.007772341	0.015643805
88	0.9126134	0.8250672	0.001849585	0.003742422
37	0.9327536	0.8653510	0.007193553	0.014407185
40	0.9322427	0.8643107	0.007338364	0.014749961
43	0.9322424	0.8643655	0.006499207	0.013037333
46	0.9354838	0.8707722	0.005869031	0.011829889
49	0.9330943	0.8660246	0.009998644	0.020027952
52	0.9365078	0.8728288	0.004031065	0.008073719
91	0.9296823	0.8591787	0.009008923	0.018080823
94	0.9354844	0.8707998	0.008497800	0.017032631
97	0.9383864	0.8766016	0.005225076	0.010501251
100	0.9346323	0.8691328	0.011350249	0.022797375
103	0.9358263	0.8714964	0.006240441	0.012489735
106	0.9378732	0.8756136	0.008037648	0.016114799

2	0.8692610	0.7382759	0.004108904	0.008277663
5	0.8677246	0.7351622	0.010678752	0.021611313
8	0.8644822	0.7286774	0.005792185	0.011706117
11	0.8719924	0.7436955	0.009996190	0.020251192
14	0.8648238	0.7293116	0.007823747	0.015864160
17	0.8622637	0.7241913	0.010044161	0.020225345
56	0.8815501	0.7629622	0.004050889	0.008241911
59	0.8760885	0.7518962	0.008382163	0.016917368
62	0.8740400	0.7478564	0.012089739	0.024324609
65	0.8805277	0.7608782	0.012059098	0.024244219
68	0.8769413	0.7536688	0.007475133	0.015140117
71	0.8735291	0.7467536	0.007521740	0.015231507
20	0.9276328	0.8551522	0.003637289	0.007335108
23	0.9283160	0.8564155	0.004816422	0.009671040
26	0.9272917	0.8544346	0.004721680	0.009454984
29	0.9231951	0.8462029	0.004474537	0.009044438
32	0.9235372	0.8469297	0.005197092	0.010343115
35	0.9298517	0.8595666	0.002361409	0.004709649
74	0.9276332	0.8551348	0.007846561	0.015756603
77	0.9339483	0.8677307	0.004109172	0.008214548
80	0.9370193	0.8738846	0.007812680	0.015702264
83	0.9293403	0.8585301	0.008360476	0.016713556
86	0.9289973	0.8577944	0.007031957	0.014176799
89	0.9366785	0.8732398	0.002371713	0.004755585
38	0.9447009	0.8892635	0.006560897	0.013107460
41	0.9479451	0.8957437	0.009700329	0.019432994
44	0.9499920	0.8998480	0.006397377	0.012861368
47	0.9445307	0.8889456	0.006950095	0.013921897
50	0.9498218	0.8995014	0.007845628	0.015712852
53	0.9556246	0.9110982	0.002472928	0.004966781
92	0.9423132	0.8844861	0.009575874	0.019205818
95	0.9486261	0.8971303	0.006193822	0.012417494
98	0.9508461	0.9015657	0.006130320	0.012325429
101	0.9487970	0.8974836	0.005359215	0.010767762
104	0.9489696	0.8978144	0.010394716	0.020830352
107	0.9518698	0.9036146	0.006672973	0.013387529
3	0.8880367	0.7759245	0.004921629	0.009921978
6	0.8803558	0.7605015	0.008883679	0.017971315
9	0.8801847	0.7601216	0.006618068	0.013356913
12	0.8858180	0.7713067	0.010817437	0.021796358
15	0.8882062	0.7761062	0.007462944	0.015177032
18	0.8783070	0.7563896	0.006227299	0.012654767
57	0.8981065	0.7960214	0.007676713	0.015336954

```
60  0.8931577 0.7861164 0.008471221 0.016944051
63  0.8890604 0.7779612 0.008368066 0.016791027
66  0.8943504 0.7884869 0.003745173 0.007529931
69  0.8895732 0.7789232 0.008149588 0.016337947
72  0.8928147 0.7854466 0.001948968 0.003924649
21  0.9329238 0.8657123 0.004116919 0.008250872
24  0.9395808 0.8790090 0.005915460 0.011870379
27  0.9361667 0.8721702 0.002204525 0.004417877
30  0.9361663 0.8721994 0.002864811 0.005801642
33  0.9324118 0.8646610 0.003482747 0.006924220
36  0.9382155 0.8763156 0.006393409 0.012793957
75  0.9375333 0.8749555 0.005493263 0.010909626
78  0.9429932 0.8858465 0.005483127 0.010990496
81  0.9419703 0.8838193 0.004701828 0.009406391
84  0.9358266 0.8715229 0.006742308 0.013441439
87  0.9418007 0.8834597 0.006117738 0.012234635
90  0.9453833 0.8906380 0.001590257 0.003131297
39  0.9554540 0.9107895 0.003963541 0.007922288
42  0.9549427 0.9097657 0.009209745 0.018423193
45  0.9557951 0.9114763 0.004102885 0.008217971
48  0.9496507 0.8991919 0.004258227 0.008563165
51  0.9586972 0.9172796 0.007247879 0.014515381
54  0.9605741 0.9210385 0.003149486 0.006295479
93  0.9481141 0.8961146 0.007520160 0.015077327
96  0.9525508 0.9049954 0.005688074 0.011406792
99  0.9549422 0.9097825 0.005499967 0.011042284
102 0.9542585 0.9083978 0.006814561 0.013674254
105 0.9511875 0.9022550 0.006746108 0.013522712
108 0.9578440 0.9155807 0.008409511 0.016812397

   nrounds max_depth eta gamma colsample_bytree min_child_weight subsample
54     150         3 0.3     0              0.8                 1         1

Model saved as 'Base_Models/XGBoostNoFeatureEngineeringPCA.rds'. To load it later, use:

loaded_model <- readRDS('Base_Models/XGBoostNoFeatureEngineeringPCA.rds')

BestTune hyperparameters saved as 'Base_Models_Data/XGBoost/BestHyperparams_XGBoostNoFeatureE

Confusion Matrix and Statistics
```

```
          Reference
Prediction Drowsy Natural
   Drowsy     294       1
   Natural    101     345

               Accuracy : 0.8623
                 95% CI : (0.8354, 0.8863)
    No Information Rate : 0.5331
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.7284

 Mcnemar's Test P-Value : < 2.2e-16

            Sensitivity : 0.7443
            Specificity : 0.9971
         Pos Pred Value : 0.9966
         Neg Pred Value : 0.7735
             Prevalence : 0.5331
         Detection Rate : 0.3968
   Detection Prevalence : 0.3981
      Balanced Accuracy : 0.8707

       'Positive' Class : Drowsy


Confusion matrix saved as 'Base_Models_Data/XGBoost/ConfusionMatrix_XGBoostNoFeatureEngineeri

Performance metrics saved as 'Base_Models_Data/XGBoost/Metrics_XGBoostNoFeatureEngineeringPCA
ROC Data for plot building saved as 'Base_Models_Data/XGBoost/ROCData_XGBoostNoFeatureEngine

ROC plot saved as 'Base_Models_Data/XGBoost/ROC_XGBoostNoFeatureEngineeringPCA.png'.
```

RandomForest, No feature pre-processing, PCA

```
  mtry  Accuracy     Kappa  AccuracySD     KappaSD
1    2 0.9534052 0.9065086 0.007548384 0.015175911
2   92 0.9624511 0.9247781 0.001344204 0.002727802
3  183 0.9547717 0.9094095 0.004419696 0.008850338

  mtry
2   92
```

```
Model saved as 'Base_Models/RandomForestNoFeatureEngineeringPCA.rds'. To load it later, use:

loaded_model <- readRDS('Base_Models/RandomForestNoFeatureEngineeringPCA.rds')

BestTune hyperparameters saved as 'Base_Models_Data/Random_Forest/BestHyperparams_RandomFores

Confusion Matrix and Statistics

          Reference
Prediction Drowsy Natural
   Drowsy     212       2
   Natural    183     344

               Accuracy : 0.7503
                 95% CI : (0.7175, 0.7811)
    No Information Rate : 0.5331
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.5142

 Mcnemar's Test P-Value : < 2.2e-16

            Sensitivity : 0.5367
            Specificity : 0.9942
         Pos Pred Value : 0.9907
         Neg Pred Value : 0.6528
             Prevalence : 0.5331
         Detection Rate : 0.2861
   Detection Prevalence : 0.2888
      Balanced Accuracy : 0.7655

       'Positive' Class : Drowsy


Confusion matrix saved as 'Base_Models_Data/Random_Forest/ConfusionMatrix_RandomForestNoFeatu

Performance metrics saved as 'Base_Models_Data/Random_Forest/Metrics_RandomForestNoFeatureEng
ROC data saved as 'Base_Models_Data/Random_Forest/ROCData_RandomForestNoFeatureEngineeringPCA

ROC plot saved as 'Base_Models_Data/Random_Forest/ROC_RandomForestNoFeatureEngineeringPCA.png
```

Gaussian SVM, feature pre-processing, PCA

```
         sigma    C  Accuracy     Kappa AccuracySD    KappaSD
1 0.0004426479 0.25 0.7970624 0.5988513 0.01038448 0.02022610
2 0.0004426479 0.50 0.8750642 0.7504102 0.01158075 0.02326455
3 0.0004426479 1.00 0.8974229 0.7945862 0.01143603 0.02296640

         sigma C
3 0.0004426479 1

Model saved as 'Base_Models/SVMModelWithFeatureEngineeringPCA.rds'. To load it later, use:

loaded_model <- readRDS('Base_Models/SVMModelWithFeatureEngineeringPCA.rds')

BestTune hyperparameters saved as 'Base_Models_Data/SVM/BestHyperparams_SVMWithFeatureEngine

Confusion Matrix and Statistics

          Reference
Prediction Drowsy Natural
   Drowsy     290      84
   Natural    105     262

               Accuracy : 0.7449
                 95% CI : (0.7119, 0.776)
    No Information Rate : 0.5331
    P-Value [Acc > NIR] : <2e-16

                  Kappa : 0.4896

 Mcnemar's Test P-Value : 0.1457

            Sensitivity : 0.7342
            Specificity : 0.7572
         Pos Pred Value : 0.7754
         Neg Pred Value : 0.7139
             Prevalence : 0.5331
         Detection Rate : 0.3914
   Detection Prevalence : 0.5047
      Balanced Accuracy : 0.7457
```

```
       'Positive' Class : Drowsy


Confusion matrix saved as 'Base_Models_Data/SVM/ConfusionMatrix_SVMWithFeatureEngineeringPCA

Performance metrics saved as 'Base_Models_Data/SVM/Metrics_SVMWithFeatureEngineeringPCA.csv'
ROC Data for plot building saved as 'Base_Models_Data/SVM/ROCData_SVMWithFeatureEngineeringP

ROC plot saved as 'Base_Models_Data/SVM/ROC_SVMWithFeatureEngineeringPCA.png'.
```

XGBoost, feature pre-processing, PCA

```
    eta max_depth gamma colsample_bytree min_child_weight subsample nrounds
1   0.3         1     0              0.6                1      0.50      50
4   0.3         1     0              0.6                1      0.75      50
7   0.3         1     0              0.6                1      1.00      50
10  0.3         1     0              0.8                1      0.50      50
13  0.3         1     0              0.8                1      0.75      50
16  0.3         1     0              0.8                1      1.00      50
55  0.4         1     0              0.6                1      0.50      50
58  0.4         1     0              0.6                1      0.75      50
61  0.4         1     0              0.6                1      1.00      50
64  0.4         1     0              0.8                1      0.50      50
67  0.4         1     0              0.8                1      0.75      50
70  0.4         1     0              0.8                1      1.00      50
19  0.3         2     0              0.6                1      0.50      50
22  0.3         2     0              0.6                1      0.75      50
25  0.3         2     0              0.6                1      1.00      50
28  0.3         2     0              0.8                1      0.50      50
31  0.3         2     0              0.8                1      0.75      50
34  0.3         2     0              0.8                1      1.00      50
73  0.4         2     0              0.6                1      0.50      50
76  0.4         2     0              0.6                1      0.75      50
79  0.4         2     0              0.6                1      1.00      50
82  0.4         2     0              0.8                1      0.50      50
85  0.4         2     0              0.8                1      0.75      50
88  0.4         2     0              0.8                1      1.00      50
37  0.3         3     0              0.6                1      0.50      50
40  0.3         3     0              0.6                1      0.75      50
43  0.3         3     0              0.6                1      1.00      50
46  0.3         3     0              0.8                1      0.50      50
```

49	0.3	3	0	0.8	1	0.75	50
52	0.3	3	0	0.8	1	1.00	50
91	0.4	3	0	0.6	1	0.50	50
94	0.4	3	0	0.6	1	0.75	50
97	0.4	3	0	0.6	1	1.00	50
100	0.4	3	0	0.8	1	0.50	50
103	0.4	3	0	0.8	1	0.75	50
106	0.4	3	0	0.8	1	1.00	50
2	0.3	1	0	0.6	1	0.50	100
5	0.3	1	0	0.6	1	0.75	100
8	0.3	1	0	0.6	1	1.00	100
11	0.3	1	0	0.8	1	0.50	100
14	0.3	1	0	0.8	1	0.75	100
17	0.3	1	0	0.8	1	1.00	100
56	0.4	1	0	0.6	1	0.50	100
59	0.4	1	0	0.6	1	0.75	100
62	0.4	1	0	0.6	1	1.00	100
65	0.4	1	0	0.8	1	0.50	100
68	0.4	1	0	0.8	1	0.75	100
71	0.4	1	0	0.8	1	1.00	100
20	0.3	2	0	0.6	1	0.50	100
23	0.3	2	0	0.6	1	0.75	100
26	0.3	2	0	0.6	1	1.00	100
29	0.3	2	0	0.8	1	0.50	100
32	0.3	2	0	0.8	1	0.75	100
35	0.3	2	0	0.8	1	1.00	100
74	0.4	2	0	0.6	1	0.50	100
77	0.4	2	0	0.6	1	0.75	100
80	0.4	2	0	0.6	1	1.00	100
83	0.4	2	0	0.8	1	0.50	100
86	0.4	2	0	0.8	1	0.75	100
89	0.4	2	0	0.8	1	1.00	100
38	0.3	3	0	0.6	1	0.50	100
41	0.3	3	0	0.6	1	0.75	100
44	0.3	3	0	0.6	1	1.00	100
47	0.3	3	0	0.8	1	0.50	100
50	0.3	3	0	0.8	1	0.75	100
53	0.3	3	0	0.8	1	1.00	100
92	0.4	3	0	0.6	1	0.50	100
95	0.4	3	0	0.6	1	0.75	100
98	0.4	3	0	0.6	1	1.00	100
101	0.4	3	0	0.8	1	0.50	100
104	0.4	3	0	0.8	1	0.75	100

107	0.4	3	0	0.8	1	1.00	100
3	0.3	1	0	0.6	1	0.50	150
6	0.3	1	0	0.6	1	0.75	150
9	0.3	1	0	0.6	1	1.00	150
12	0.3	1	0	0.8	1	0.50	150
15	0.3	1	0	0.8	1	0.75	150
18	0.3	1	0	0.8	1	1.00	150
57	0.4	1	0	0.6	1	0.50	150
60	0.4	1	0	0.6	1	0.75	150
63	0.4	1	0	0.6	1	1.00	150
66	0.4	1	0	0.8	1	0.50	150
69	0.4	1	0	0.8	1	0.75	150
72	0.4	1	0	0.8	1	1.00	150
21	0.3	2	0	0.6	1	0.50	150
24	0.3	2	0	0.6	1	0.75	150
27	0.3	2	0	0.6	1	1.00	150
30	0.3	2	0	0.8	1	0.50	150
33	0.3	2	0	0.8	1	0.75	150
36	0.3	2	0	0.8	1	1.00	150
75	0.4	2	0	0.6	1	0.50	150
78	0.4	2	0	0.6	1	0.75	150
81	0.4	2	0	0.6	1	1.00	150
84	0.4	2	0	0.8	1	0.50	150
87	0.4	2	0	0.8	1	0.75	150
90	0.4	2	0	0.8	1	1.00	150
39	0.3	3	0	0.6	1	0.50	150
42	0.3	3	0	0.6	1	0.75	150
45	0.3	3	0	0.6	1	1.00	150
48	0.3	3	0	0.8	1	0.50	150
51	0.3	3	0	0.8	1	0.75	150
54	0.3	3	0	0.8	1	1.00	150
93	0.4	3	0	0.6	1	0.50	150
96	0.4	3	0	0.6	1	0.75	150
99	0.4	3	0	0.6	1	1.00	150
102	0.4	3	0	0.8	1	0.50	150
105	0.4	3	0	0.8	1	0.75	150
108	0.4	3	0	0.8	1	1.00	150

	Accuracy	Kappa	AccuracySD	KappaSD
1	0.8047438	0.6086276	0.007361343	0.014960645
4	0.8025259	0.6039578	0.003944092	0.007674354
7	0.8078163	0.6144657	0.005922455	0.012022202
10	0.8122514	0.6234408	0.012562724	0.025591520
13	0.8115693	0.6221969	0.008992123	0.018150997

16	0.8149847	0.6288576	0.007521131	0.015182657
55	0.8139598	0.6269251	0.010693929	0.021610136
58	0.8136175	0.6259536	0.009427523	0.018845925
61	0.8131063	0.6251494	0.006755793	0.013629500
64	0.8107173	0.6200285	0.015358670	0.031023842
67	0.8182259	0.6352089	0.009037010	0.018490039
70	0.8161810	0.6311720	0.009532753	0.019375511
19	0.8412672	0.6815355	0.009192630	0.018696471
22	0.8503130	0.6996989	0.012360128	0.024937870
25	0.8470719	0.6929576	0.005065493	0.010256479
28	0.8530459	0.7052882	0.003725782	0.007603659
31	0.8446818	0.6883166	0.006028018	0.012187358
34	0.8462185	0.6913306	0.012489091	0.025189862
73	0.8498037	0.6988434	0.003076943	0.006163964
76	0.8542391	0.7075779	0.012824553	0.025877618
79	0.8559469	0.7109905	0.007765175	0.015776054
82	0.8431451	0.6854275	0.013052023	0.026290715
85	0.8550915	0.7093306	0.013342412	0.026730471
88	0.8535587	0.7060663	0.008106272	0.016467986
37	0.8518518	0.7028674	0.011233483	0.022632412
40	0.8585081	0.7161959	0.010886437	0.022001167
43	0.8685763	0.7364000	0.012049493	0.024205124
46	0.8559487	0.7110475	0.004086332	0.008337927
49	0.8634570	0.7261911	0.009186900	0.018509765
52	0.8707952	0.7408009	0.010543198	0.021332601
91	0.8504860	0.7003611	0.009245676	0.018544612
94	0.8603851	0.7202582	0.003681792	0.007334104
97	0.8682372	0.7357850	0.007017761	0.014003734
100	0.8559481	0.7112156	0.005769243	0.011734928
103	0.8670418	0.7335327	0.004078825	0.008100407
106	0.8709662	0.7412810	0.006060476	0.012201733
2	0.8346122	0.6681501	0.012696212	0.025846648
5	0.8279540	0.6545234	0.010934856	0.022212248
8	0.8288084	0.6562177	0.007891043	0.016115243
11	0.8322214	0.6635437	0.009420375	0.018968770
14	0.8342708	0.6674510	0.006850892	0.014020365
17	0.8311976	0.6609936	0.010453142	0.021078632
56	0.8375154	0.6741808	0.008533858	0.017206426
59	0.8402442	0.6794605	0.008643424	0.017494362
62	0.8390489	0.6769215	0.008766174	0.017773247
65	0.8320517	0.6631636	0.012744704	0.025744801
68	0.8390499	0.6771284	0.005524696	0.011210447
71	0.8330759	0.6649139	0.007346259	0.014821196

20	0.8559482	0.7110653	0.006939474	0.013935181
23	0.8605554	0.7203387	0.008734105	0.017578029
26	0.8682355	0.7357088	0.008866176	0.017862897
29	0.8624330	0.7240724	0.011096010	0.022390765
32	0.8666995	0.7326461	0.006627005	0.013213732
35	0.8667014	0.7326777	0.011461137	0.023192360
74	0.8610685	0.7216587	0.006278640	0.012521302
77	0.8653333	0.7299620	0.011455304	0.023090799
80	0.8711381	0.7415428	0.003032802	0.006207282
83	0.8545832	0.7084121	0.011992885	0.024147901
86	0.8651650	0.7297581	0.008244738	0.016681014
89	0.8711364	0.7415472	0.009009298	0.018215306
38	0.8629460	0.7252135	0.014748084	0.029739695
41	0.8748937	0.7491945	0.004723231	0.009582440
44	0.8776244	0.7545639	0.010405291	0.020825622
47	0.8668709	0.7330103	0.008712979	0.017374368
50	0.8760856	0.7515416	0.009104776	0.018366173
53	0.8861572	0.7717747	0.005857941	0.011804499
92	0.8615808	0.7226576	0.003868251	0.007689618
95	0.8719918	0.7434980	0.009728582	0.019619412
98	0.8777938	0.7549620	0.007145812	0.014411194
101	0.8631154	0.7256835	0.009908245	0.019957099
104	0.8745520	0.7485085	0.007843481	0.015648156
107	0.8818894	0.7632144	0.006801697	0.013708588
3	0.8424626	0.6841650	0.012739871	0.025408772
6	0.8399030	0.6786783	0.008912048	0.018024129
9	0.8443387	0.6874234	0.012620615	0.025424297
12	0.8463884	0.6920352	0.008071562	0.016278425
15	0.8462179	0.6914537	0.007319313	0.014810236
18	0.8436574	0.6861131	0.010301883	0.020848461
57	0.8455365	0.6904241	0.001842126	0.003636039
60	0.8496304	0.6984746	0.010191870	0.020456730
63	0.8527025	0.7043111	0.011693029	0.023712927
66	0.8378554	0.6748201	0.006718856	0.013525101
69	0.8429766	0.6850720	0.007010357	0.014183755
72	0.8472408	0.6933994	0.011918303	0.024046795
21	0.8697722	0.7388287	0.006479472	0.013108648
24	0.8702828	0.7398278	0.013343730	0.026902471
27	0.8733555	0.7460723	0.007588401	0.015270722
30	0.8680661	0.7354473	0.017346859	0.034726238
33	0.8690887	0.7375426	0.005758355	0.011426010
36	0.8764301	0.7521909	0.009744010	0.019647673
75	0.8607275	0.7209422	0.008467548	0.016992045

```
78  0.8661877 0.7316773 0.010012722 0.020095204
81  0.8735276 0.7462781 0.006259483 0.012590372
84  0.8537288 0.7068031 0.013506410 0.027270527
87  0.8747220 0.7488666 0.007086234 0.014283107
90  0.8745514 0.7485383 0.010123109 0.020410356
39  0.8690880 0.7375466 0.012523489 0.025169516
42  0.8818919 0.7631778 0.005786714 0.011767961
45  0.8868408 0.7731182 0.008980096 0.017953474
48  0.8689197 0.7372109 0.007426847 0.015028363
51  0.8858150 0.7709185 0.009649431 0.019431463
54  0.8938384 0.7871721 0.006231258 0.012509273
93  0.8668718 0.7332141 0.004381709 0.008839012
96  0.8803540 0.7601977 0.007637828 0.015439191
99  0.8844502 0.7682290 0.009530645 0.019242764
102 0.8661874 0.7319081 0.008155323 0.016150528
105 0.8806966 0.7607895 0.003629713 0.007330050
108 0.8878638 0.7752352 0.006592183 0.013237102

   nrounds max_depth eta gamma colsample_bytree min_child_weight subsample
54     150         3 0.3     0              0.8                1         1

Model saved as 'Base_Models/XGBoostWithFeatureEngineeringPCA.rds'. To load it later, use:

loaded_model <- readRDS('Base_Models/XGBoostWithFeatureEngineeringPCA.rds')

BestTune hyperparameters saved as 'Base_Models_Data/XGBoost/BestHyperparams_XGBoostWithFeatu

Confusion Matrix and Statistics

          Reference
Prediction Drowsy Natural
   Drowsy     101      19
   Natural    294     327

               Accuracy : 0.5776
                 95% CI : (0.5411, 0.6135)
    No Information Rate : 0.5331
    P-Value [Acc > NIR] : 0.008237

                  Kappa : 0.1913
```

```
 Mcnemar's Test P-Value : < 2.2e-16

            Sensitivity : 0.2557
            Specificity : 0.9451
         Pos Pred Value : 0.8417
         Neg Pred Value : 0.5266
             Prevalence : 0.5331
         Detection Rate : 0.1363
   Detection Prevalence : 0.1619
      Balanced Accuracy : 0.6004

       'Positive' Class : Drowsy


Confusion matrix saved as 'Base_Models_Data/XGBoost/ConfusionMatrix_XGBoostWithFeatureEngine

Performance metrics saved as 'Base_Models_Data/XGBoost/Metrics_XGBoostWithFeatureEngineering
ROC Data for plot building saved as 'Base_Models_Data/XGBoost/ROCData_XGBoostWithFeatureEngi

ROC plot saved as 'Base_Models_Data/XGBoost/ROC_XGBoostWithFeatureEngineeringPCA.png'.
```

RandomForest, feature pre-processing, PCA

```
  mtry  Accuracy     Kappa  AccuracySD     KappaSD
1    2 0.7869947 0.5682807 0.002959987 0.006315773
2   51 0.8752335 0.7490753 0.012917997 0.026202362
3 1312 0.8774520 0.7543106 0.006553174 0.013253912

  mtry
3 1312

Model saved as 'Base_Models/RandomForestWithFeatureEngineeringPCA.rds'. To load it later, us

loaded_model <- readRDS('Base_Models/RandomForestWithFeatureEngineeringPCA.rds')

BestTune hyperparameters saved as 'Base_Models_Data/Random_Forest/BestHyperparams_RandomFore
```

```
Confusion Matrix and Statistics

          Reference
Prediction Drowsy Natural
   Drowsy     127      57
   Natural    268     289

               Accuracy : 0.5614
                 95% CI : (0.5248, 0.5975)
    No Information Rate : 0.5331
    P-Value [Acc > NIR] : 0.06543

                  Kappa : 0.1511

 Mcnemar's Test P-Value : < 2e-16

            Sensitivity : 0.3215
            Specificity : 0.8353
         Pos Pred Value : 0.6902
         Neg Pred Value : 0.5189
             Prevalence : 0.5331
         Detection Rate : 0.1714
   Detection Prevalence : 0.2483
      Balanced Accuracy : 0.5784

       'Positive' Class : Drowsy


Confusion matrix saved as 'Base_Models_Data/Random_Forest/ConfusionMatrix_RandomForestWithFea

Performance metrics saved as 'Base_Models_Data/Random_Forest/Metrics_RandomForestWithFeatureE
ROC data saved as 'Base_Models_Data/Random_Forest/ROCData_RandomForestWithFeatureEngineeringP

ROC plot saved as 'Base_Models_Data/Random_Forest/ROC_RandomForestWithFeatureEngineeringPCA.p
```

Gaussian SVM, No feature pre-processing

```
         sigma    C  Accuracy     Kappa  AccuracySD    KappaSD
1 0.0002832271 0.25 0.9143193 0.8284436 0.010431265 0.02093534
2 0.0002832271 0.50 0.9378730 0.8755054 0.010632073 0.02137275
3 0.0002832271 1.00 0.9598913 0.9196224 0.006795304 0.01366191
```

```
         sigma C
3 0.0002832271 1

Model saved as 'Base_Models/SVMModelNoFeatureEngineering.rds'. To load it later, use:

loaded_model <- readRDS('Base_Models/SVMModelNoFeatureEngineering.rds')

BestTune hyperparameters saved as 'Base_Models_Data/SVM/BestHyperparams_SVMNoFeatureEngineeri

Confusion Matrix and Statistics

          Reference
Prediction Drowsy Natural
   Drowsy     291       0
   Natural    104     346

               Accuracy : 0.8596
                 95% CI : (0.8325, 0.8839)
    No Information Rate : 0.5331
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.7232

 Mcnemar's Test P-Value : < 2.2e-16

            Sensitivity : 0.7367
            Specificity : 1.0000
         Pos Pred Value : 1.0000
         Neg Pred Value : 0.7689
             Prevalence : 0.5331
         Detection Rate : 0.3927
   Detection Prevalence : 0.3927
      Balanced Accuracy : 0.8684

       'Positive' Class : Drowsy


Confusion matrix saved as 'Base_Models_Data/SVM/ConfusionMatrix_SVMNoFeatureEngineering.csv'

Performance metrics saved as 'Base_Models_Data/SVM/Metrics_SVMNoFeatureEngineering.csv'.
ROC Data for plot building saved as 'Base_Models_Data/SVM/ROCData_SVMNoFeatureEngineering.cs
```

```
ROC plot saved as 'Base_Models_Data/SVM/ROC_SVMNoFeatureEngineering.png'.
```

XGBoost, No feature pre-processing

	eta	max_depth	gamma	colsample_bytree	min_child_weight	subsample	nrounds
1	0.3	1	0	0.6	1	0.50	50
4	0.3	1	0	0.6	1	0.75	50
7	0.3	1	0	0.6	1	1.00	50
10	0.3	1	0	0.8	1	0.50	50
13	0.3	1	0	0.8	1	0.75	50
16	0.3	1	0	0.8	1	1.00	50
55	0.4	1	0	0.6	1	0.50	50
58	0.4	1	0	0.6	1	0.75	50
61	0.4	1	0	0.6	1	1.00	50
64	0.4	1	0	0.8	1	0.50	50
67	0.4	1	0	0.8	1	0.75	50
70	0.4	1	0	0.8	1	1.00	50
19	0.3	2	0	0.6	1	0.50	50
22	0.3	2	0	0.6	1	0.75	50
25	0.3	2	0	0.6	1	1.00	50
28	0.3	2	0	0.8	1	0.50	50
31	0.3	2	0	0.8	1	0.75	50
34	0.3	2	0	0.8	1	1.00	50
73	0.4	2	0	0.6	1	0.50	50
76	0.4	2	0	0.6	1	0.75	50
79	0.4	2	0	0.6	1	1.00	50
82	0.4	2	0	0.8	1	0.50	50
85	0.4	2	0	0.8	1	0.75	50
88	0.4	2	0	0.8	1	1.00	50
37	0.3	3	0	0.6	1	0.50	50
40	0.3	3	0	0.6	1	0.75	50
43	0.3	3	0	0.6	1	1.00	50
46	0.3	3	0	0.8	1	0.50	50
49	0.3	3	0	0.8	1	0.75	50
52	0.3	3	0	0.8	1	1.00	50
91	0.4	3	0	0.6	1	0.50	50
94	0.4	3	0	0.6	1	0.75	50
97	0.4	3	0	0.6	1	1.00	50
100	0.4	3	0	0.8	1	0.50	50
103	0.4	3	0	0.8	1	0.75	50
106	0.4	3	0	0.8	1	1.00	50
2	0.3	1	0	0.6	1	0.50	100

5	0.3	1	0	0.6	1	0.75	100
8	0.3	1	0	0.6	1	1.00	100
11	0.3	1	0	0.8	1	0.50	100
14	0.3	1	0	0.8	1	0.75	100
17	0.3	1	0	0.8	1	1.00	100
56	0.4	1	0	0.6	1	0.50	100
59	0.4	1	0	0.6	1	0.75	100
62	0.4	1	0	0.6	1	1.00	100
65	0.4	1	0	0.8	1	0.50	100
68	0.4	1	0	0.8	1	0.75	100
71	0.4	1	0	0.8	1	1.00	100
20	0.3	2	0	0.6	1	0.50	100
23	0.3	2	0	0.6	1	0.75	100
26	0.3	2	0	0.6	1	1.00	100
29	0.3	2	0	0.8	1	0.50	100
32	0.3	2	0	0.8	1	0.75	100
35	0.3	2	0	0.8	1	1.00	100
74	0.4	2	0	0.6	1	0.50	100
77	0.4	2	0	0.6	1	0.75	100
80	0.4	2	0	0.6	1	1.00	100
83	0.4	2	0	0.8	1	0.50	100
86	0.4	2	0	0.8	1	0.75	100
89	0.4	2	0	0.8	1	1.00	100
38	0.3	3	0	0.6	1	0.50	100
41	0.3	3	0	0.6	1	0.75	100
44	0.3	3	0	0.6	1	1.00	100
47	0.3	3	0	0.8	1	0.50	100
50	0.3	3	0	0.8	1	0.75	100
53	0.3	3	0	0.8	1	1.00	100
92	0.4	3	0	0.6	1	0.50	100
95	0.4	3	0	0.6	1	0.75	100
98	0.4	3	0	0.6	1	1.00	100
101	0.4	3	0	0.8	1	0.50	100
104	0.4	3	0	0.8	1	0.75	100
107	0.4	3	0	0.8	1	1.00	100
3	0.3	1	0	0.6	1	0.50	150
6	0.3	1	0	0.6	1	0.75	150
9	0.3	1	0	0.6	1	1.00	150
12	0.3	1	0	0.8	1	0.50	150
15	0.3	1	0	0.8	1	0.75	150
18	0.3	1	0	0.8	1	1.00	150
57	0.4	1	0	0.6	1	0.50	150
60	0.4	1	0	0.6	1	0.75	150

63	0.4	1	0	0.6	1	1.00	150
66	0.4	1	0	0.8	1	0.50	150
69	0.4	1	0	0.8	1	0.75	150
72	0.4	1	0	0.8	1	1.00	150
21	0.3	2	0	0.6	1	0.50	150
24	0.3	2	0	0.6	1	0.75	150
27	0.3	2	0	0.6	1	1.00	150
30	0.3	2	0	0.8	1	0.50	150
33	0.3	2	0	0.8	1	0.75	150
36	0.3	2	0	0.8	1	1.00	150
75	0.4	2	0	0.6	1	0.50	150
78	0.4	2	0	0.6	1	0.75	150
81	0.4	2	0	0.6	1	1.00	150
84	0.4	2	0	0.8	1	0.50	150
87	0.4	2	0	0.8	1	0.75	150
90	0.4	2	0	0.8	1	1.00	150
39	0.3	3	0	0.6	1	0.50	150
42	0.3	3	0	0.6	1	0.75	150
45	0.3	3	0	0.6	1	1.00	150
48	0.3	3	0	0.8	1	0.50	150
51	0.3	3	0	0.8	1	0.75	150
54	0.3	3	0	0.8	1	1.00	150
93	0.4	3	0	0.6	1	0.50	150
96	0.4	3	0	0.6	1	0.75	150
99	0.4	3	0	0.6	1	1.00	150
102	0.4	3	0	0.8	1	0.50	150
105	0.4	3	0	0.8	1	0.75	150
108	0.4	3	0	0.8	1	1.00	150

	Accuracy	Kappa	AccuracySD	KappaSD
1	0.8605551	0.7211344	0.012754072	0.025654441
4	0.8600420	0.7200171	0.014385029	0.028813086
7	0.8574840	0.7148530	0.010470081	0.021197656
10	0.8564605	0.7129920	0.013274654	0.026834787
13	0.8559463	0.7119183	0.016158699	0.032385315
16	0.8636267	0.7271898	0.015863871	0.031629550
55	0.8702873	0.7404509	0.016797686	0.033842054
58	0.8709676	0.7417453	0.016736502	0.033595663
61	0.8711388	0.7422605	0.011473289	0.023264206
64	0.8716501	0.7432136	0.008093365	0.016314567
67	0.8678948	0.7355636	0.012997852	0.026149880
70	0.8711371	0.7421355	0.010117979	0.020425472
19	0.9218303	0.8434688	0.011339787	0.022724264
22	0.9180743	0.8359009	0.015678040	0.031568276

25	0.9218292	0.8434572	0.013676898	0.027473731
28	0.9179037	0.8355829	0.008900744	0.017816591
31	0.9192700	0.8383533	0.010420787	0.020968262
34	0.9219995	0.8437974	0.014975152	0.030005863
73	0.9228548	0.8455249	0.010861301	0.021804357
76	0.9276316	0.8550408	0.007749210	0.015546649
79	0.9298517	0.8595170	0.013414083	0.026901747
82	0.9254123	0.8506347	0.009681385	0.019414145
85	0.9303641	0.8605311	0.008624967	0.017365804
88	0.9272898	0.8543718	0.010244028	0.020539400
37	0.9489663	0.8977575	0.009649814	0.019346423
40	0.9506734	0.9012074	0.010017532	0.020120308
43	0.9472616	0.8943250	0.004558323	0.009230132
46	0.9462367	0.8922747	0.008742053	0.017545443
49	0.9470896	0.8940126	0.005158783	0.010326130
52	0.9472620	0.8943326	0.009876360	0.019828581
91	0.9484557	0.8967666	0.009079759	0.018252885
94	0.9474317	0.8946891	0.006996327	0.014035251
97	0.9518700	0.9035835	0.010489868	0.021023659
100	0.9505041	0.9008462	0.009928757	0.019911056
103	0.9493082	0.8984247	0.007107945	0.014255836
106	0.9527219	0.9053053	0.005618773	0.011267722
2	0.8899124	0.7797230	0.017470222	0.035121535
5	0.8894007	0.7786058	0.011757444	0.023561208
8	0.8900839	0.7800551	0.010694661	0.021491303
11	0.8921315	0.7840285	0.014350964	0.028827468
14	0.8904247	0.7807346	0.010341187	0.020798141
17	0.8938380	0.7875199	0.009385632	0.018781139
56	0.9023736	0.8045450	0.012243996	0.024491356
59	0.8994714	0.7987461	0.012898273	0.025887080
62	0.8994714	0.7987429	0.010977251	0.022167029
65	0.9001526	0.8000972	0.011803548	0.023727689
68	0.8974219	0.7946466	0.006984838	0.014138324
71	0.8965709	0.7929862	0.010884054	0.021837012
20	0.9399215	0.8796500	0.010529465	0.021100928
23	0.9409460	0.8816687	0.011963980	0.024072400
26	0.9447001	0.8892531	0.012431117	0.024926492
29	0.9388976	0.8776004	0.007571914	0.015157432
32	0.9406043	0.8810634	0.010919602	0.021940190
35	0.9433352	0.8864576	0.011182416	0.022427996
74	0.9474328	0.8947061	0.008620262	0.017312790
77	0.9476017	0.8950120	0.008374982	0.016851094
80	0.9516982	0.9032414	0.009221738	0.018478873

83	0.9479426	0.8957098	0.010814372	0.021695151
86	0.9465783	0.8929807	0.008732047	0.017524115
89	0.9494786	0.8987928	0.009648114	0.019342663
38	0.9653519	0.9305910	0.004211610	0.008424440
41	0.9672295	0.9343566	0.006261064	0.012560804
44	0.9641580	0.9281881	0.005560239	0.011168453
47	0.9585251	0.9168975	0.006023084	0.012086478
50	0.9650108	0.9299041	0.005757773	0.011525170
53	0.9651816	0.9302305	0.005986197	0.011990261
92	0.9597197	0.9193215	0.012706155	0.025454790
95	0.9638174	0.9275003	0.006222798	0.012500630
98	0.9682546	0.9364025	0.005021893	0.010067648
101	0.9612558	0.9223920	0.007054288	0.014144723
104	0.9636462	0.9271742	0.007547230	0.015140740
107	0.9668882	0.9336649	0.004844640	0.009718152
3	0.9073217	0.8144460	0.016276937	0.032688753
6	0.9057860	0.8113098	0.013512055	0.027134307
9	0.9028844	0.8055735	0.007712840	0.015542787
12	0.9080043	0.8157272	0.014119896	0.028425697
15	0.9081752	0.8161547	0.006085412	0.012242165
18	0.9064695	0.8126955	0.009804972	0.019711879
57	0.9134669	0.8267782	0.015111284	0.030221962
60	0.9160283	0.8318018	0.012795154	0.025749788
63	0.9112487	0.8222654	0.008148864	0.016406182
66	0.9134669	0.8266550	0.015314669	0.030852067
69	0.9150016	0.8297683	0.007893820	0.015865705
72	0.9100530	0.8198907	0.007971479	0.016027331
21	0.9546005	0.9090729	0.007418487	0.014884646
24	0.9559661	0.9117830	0.008708653	0.017504685
27	0.9545996	0.9090450	0.012861588	0.025773263
30	0.9522110	0.9042803	0.006492019	0.012964837
33	0.9511864	0.9022376	0.008645655	0.017363705
36	0.9566484	0.9131632	0.009214387	0.018472028
75	0.9564784	0.9128240	0.008568753	0.017193004
78	0.9581841	0.9162235	0.007582816	0.015209657
81	0.9590374	0.9179284	0.006033355	0.012114822
84	0.9557945	0.9114419	0.008625973	0.017309664
87	0.9575015	0.9148715	0.008367147	0.016771338
90	0.9619388	0.9237460	0.006111740	0.012261412
39	0.9709848	0.9418661	0.003518026	0.007054611
42	0.9696189	0.9391395	0.007600892	0.015233665
45	0.9701316	0.9401567	0.002557686	0.005142903
48	0.9655229	0.9309307	0.005316799	0.010672274

```
51  0.9714966 0.9429017 0.003701940 0.007426311
54  0.9708139 0.9415169 0.006426442 0.012877551
93  0.9639869 0.9278749 0.007712997 0.015453713
96  0.9706439 0.9411778 0.003744085 0.007529682
99  0.9721799 0.9442750 0.003441310 0.006902073
102 0.9636458 0.9271803 0.007645427 0.015320224
105 0.9667184 0.9333309 0.005290130 0.010612977
108 0.9691075 0.9381064 0.005639296 0.011313651

   nrounds max_depth eta gamma colsample_bytree min_child_weight subsample
99     150         3 0.4     0              0.6                1         1

Model saved as 'Base_Models/XGBoostNoFeatureEngineering.rds'. To load it later, use:

loaded_model <- readRDS('Base_Models/XGBoostNoFeatureEngineering.rds')

BestTune hyperparameters saved as 'Base_Models_Data/XGBoost/BestHyperparams_XGBoostNoFeatureE

Confusion Matrix and Statistics

          Reference
Prediction Drowsy Natural
   Drowsy     225       1
   Natural    170     345

               Accuracy : 0.7692
                 95% CI : (0.7372, 0.7991)
    No Information Rate : 0.5331
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.5501

 Mcnemar's Test P-Value : < 2.2e-16

            Sensitivity : 0.5696
            Specificity : 0.9971
         Pos Pred Value : 0.9956
         Neg Pred Value : 0.6699
             Prevalence : 0.5331
         Detection Rate : 0.3036
   Detection Prevalence : 0.3050
```

```
      Balanced Accuracy : 0.7834

       'Positive' Class : Drowsy


Confusion matrix saved as 'Base_Models_Data/XGBoost/ConfusionMatrix_XGBoostNoFeatureEngineeri

Performance metrics saved as 'Base_Models_Data/XGBoost/Metrics_XGBoostNoFeatureEngineering.cs
ROC Data for plot building saved as 'Base_Models_Data/XGBoost/ROCData_XGBoostNoFeatureEngine

ROC plot saved as 'Base_Models_Data/XGBoost/ROC_XGBoostNoFeatureEngineering.png'.
```

Random Forest, No feature pre-processing

```
  mtry  Accuracy     Kappa  AccuracySD    KappaSD
1    2 0.9592089 0.9181975 0.007636221 0.01533393
2   67 0.9643298 0.9285142 0.008384652 0.01682170
3 2303 0.9621106 0.9241025 0.009412129 0.01885017

  mtry
2   67

Model saved as 'Base_Models/RandomForestNoFeatureEngineering.rds'. To load it later, use:

loaded_model <- readRDS('Base_Models/RandomForestNoFeatureEngineering.rds')

BestTune hyperparameters saved as 'Base_Models_Data/Random_Forest/BestHyperparams_RandomFores

Confusion Matrix and Statistics

          Reference
Prediction Drowsy Natural
   Drowsy     192       1
   Natural    203     345

               Accuracy : 0.7247
                 95% CI : (0.691, 0.7566)
    No Information Rate : 0.5331
    P-Value [Acc > NIR] : < 2.2e-16
```

```
                 Kappa : 0.4663

 Mcnemar's Test P-Value : < 2.2e-16

            Sensitivity : 0.4861
            Specificity : 0.9971
         Pos Pred Value : 0.9948
         Neg Pred Value : 0.6296
             Prevalence : 0.5331
         Detection Rate : 0.2591
   Detection Prevalence : 0.2605
      Balanced Accuracy : 0.7416

       'Positive' Class : Drowsy


Confusion matrix saved as 'Base_Models_Data/Random_Forest/ConfusionMatrix_RandomForestNoFeatu

Performance metrics saved as 'Base_Models_Data/Random_Forest/Metrics_RandomForestNoFeatureEng
ROC data saved as 'Base_Models_Data/Random_Forest/ROCData_RandomForestNoFeatureEngineering.cs

ROC plot saved as 'Base_Models_Data/Random_Forest/ROC_RandomForestNoFeatureEngineering.png'.
```

Gaussian SVM, feature pre-processing

```
         sigma    C  Accuracy     Kappa  AccuracySD    KappaSD
1 0.0002349078 0.25 0.8557764 0.7108330 0.006004945 0.01224774
2 0.0002349078 0.50 0.8861563 0.7719503 0.007703850 0.01563234
3 0.0002349078 1.00 0.9068080 0.8133797 0.010056936 0.02024820

         sigma C
3 0.0002349078 1

Model saved as 'Base_Models/SVMModelWithFeatureEngineering.rds'. To load it later, use:

loaded_model <- readRDS('Base_Models/SVMModelWithFeatureEngineering.rds')

BestTune hyperparameters saved as 'Base_Models_Data/SVM/BestHyperparams_SVMWithFeatureEngine
```

```
Confusion Matrix and Statistics

          Reference
Prediction Drowsy Natural
   Drowsy     309      92
   Natural     86     254

               Accuracy : 0.7598
                 95% CI : (0.7274, 0.7901)
    No Information Rate : 0.5331
    P-Value [Acc > NIR] : <2e-16

                  Kappa : 0.5169

 Mcnemar's Test P-Value : 0.7078

            Sensitivity : 0.7823
            Specificity : 0.7341
         Pos Pred Value : 0.7706
         Neg Pred Value : 0.7471
             Prevalence : 0.5331
         Detection Rate : 0.4170
   Detection Prevalence : 0.5412
      Balanced Accuracy : 0.7582

       'Positive' Class : Drowsy


Confusion matrix saved as 'Base_Models_Data/SVM/ConfusionMatrix_SVMWithFeatureEngineering.csv

Performance metrics saved as 'Base_Models_Data/SVM/Metrics_SVMWithFeatureEngineering.csv'.
ROC Data for plot building saved as 'Base_Models_Data/SVM/ROCData_SVMWithFeatureEngineering.c

ROC plot saved as 'Base_Models_Data/SVM/ROC_SVMWithFeatureEngineering.png'.
```

XGBoost, feature pre-processing

```
    eta max_depth gamma colsample_bytree min_child_weight subsample nrounds
1   0.3         1     0              0.6                1      0.50      50
4   0.3         1     0              0.6                1      0.75      50
7   0.3         1     0              0.6                1      1.00      50
```

10	0.3	1	0	0.8	1	0.50	50
13	0.3	1	0	0.8	1	0.75	50
16	0.3	1	0	0.8	1	1.00	50
55	0.4	1	0	0.6	1	0.50	50
58	0.4	1	0	0.6	1	0.75	50
61	0.4	1	0	0.6	1	1.00	50
64	0.4	1	0	0.8	1	0.50	50
67	0.4	1	0	0.8	1	0.75	50
70	0.4	1	0	0.8	1	1.00	50
19	0.3	2	0	0.6	1	0.50	50
22	0.3	2	0	0.6	1	0.75	50
25	0.3	2	0	0.6	1	1.00	50
28	0.3	2	0	0.8	1	0.50	50
31	0.3	2	0	0.8	1	0.75	50
34	0.3	2	0	0.8	1	1.00	50
73	0.4	2	0	0.6	1	0.50	50
76	0.4	2	0	0.6	1	0.75	50
79	0.4	2	0	0.6	1	1.00	50
82	0.4	2	0	0.8	1	0.50	50
85	0.4	2	0	0.8	1	0.75	50
88	0.4	2	0	0.8	1	1.00	50
37	0.3	3	0	0.6	1	0.50	50
40	0.3	3	0	0.6	1	0.75	50
43	0.3	3	0	0.6	1	1.00	50
46	0.3	3	0	0.8	1	0.50	50
49	0.3	3	0	0.8	1	0.75	50
52	0.3	3	0	0.8	1	1.00	50
91	0.4	3	0	0.6	1	0.50	50
94	0.4	3	0	0.6	1	0.75	50
97	0.4	3	0	0.6	1	1.00	50
100	0.4	3	0	0.8	1	0.50	50
103	0.4	3	0	0.8	1	0.75	50
106	0.4	3	0	0.8	1	1.00	50
2	0.3	1	0	0.6	1	0.50	100
5	0.3	1	0	0.6	1	0.75	100
8	0.3	1	0	0.6	1	1.00	100
11	0.3	1	0	0.8	1	0.50	100
14	0.3	1	0	0.8	1	0.75	100
17	0.3	1	0	0.8	1	1.00	100
56	0.4	1	0	0.6	1	0.50	100
59	0.4	1	0	0.6	1	0.75	100
62	0.4	1	0	0.6	1	1.00	100
65	0.4	1	0	0.8	1	0.50	100

68	0.4	1	0	0.8	1	0.75	100
71	0.4	1	0	0.8	1	1.00	100
20	0.3	2	0	0.6	1	0.50	100
23	0.3	2	0	0.6	1	0.75	100
26	0.3	2	0	0.6	1	1.00	100
29	0.3	2	0	0.8	1	0.50	100
32	0.3	2	0	0.8	1	0.75	100
35	0.3	2	0	0.8	1	1.00	100
74	0.4	2	0	0.6	1	0.50	100
77	0.4	2	0	0.6	1	0.75	100
80	0.4	2	0	0.6	1	1.00	100
83	0.4	2	0	0.8	1	0.50	100
86	0.4	2	0	0.8	1	0.75	100
89	0.4	2	0	0.8	1	1.00	100
38	0.3	3	0	0.6	1	0.50	100
41	0.3	3	0	0.6	1	0.75	100
44	0.3	3	0	0.6	1	1.00	100
47	0.3	3	0	0.8	1	0.50	100
50	0.3	3	0	0.8	1	0.75	100
53	0.3	3	0	0.8	1	1.00	100
92	0.4	3	0	0.6	1	0.50	100
95	0.4	3	0	0.6	1	0.75	100
98	0.4	3	0	0.6	1	1.00	100
101	0.4	3	0	0.8	1	0.50	100
104	0.4	3	0	0.8	1	0.75	100
107	0.4	3	0	0.8	1	1.00	100
3	0.3	1	0	0.6	1	0.50	150
6	0.3	1	0	0.6	1	0.75	150
9	0.3	1	0	0.6	1	1.00	150
12	0.3	1	0	0.8	1	0.50	150
15	0.3	1	0	0.8	1	0.75	150
18	0.3	1	0	0.8	1	1.00	150
57	0.4	1	0	0.6	1	0.50	150
60	0.4	1	0	0.6	1	0.75	150
63	0.4	1	0	0.6	1	1.00	150
66	0.4	1	0	0.8	1	0.50	150
69	0.4	1	0	0.8	1	0.75	150
72	0.4	1	0	0.8	1	1.00	150
21	0.3	2	0	0.6	1	0.50	150
24	0.3	2	0	0.6	1	0.75	150
27	0.3	2	0	0.6	1	1.00	150
30	0.3	2	0	0.8	1	0.50	150
33	0.3	2	0	0.8	1	0.75	150

36	0.3	2	0	0.8	1	1.00	150
75	0.4	2	0	0.6	1	0.50	150
78	0.4	2	0	0.6	1	0.75	150
81	0.4	2	0	0.6	1	1.00	150
84	0.4	2	0	0.8	1	0.50	150
87	0.4	2	0	0.8	1	0.75	150
90	0.4	2	0	0.8	1	1.00	150
39	0.3	3	0	0.6	1	0.50	150
42	0.3	3	0	0.6	1	0.75	150
45	0.3	3	0	0.6	1	1.00	150
48	0.3	3	0	0.8	1	0.50	150
51	0.3	3	0	0.8	1	0.75	150
54	0.3	3	0	0.8	1	1.00	150
93	0.4	3	0	0.6	1	0.50	150
96	0.4	3	0	0.6	1	0.75	150
99	0.4	3	0	0.6	1	1.00	150
102	0.4	3	0	0.8	1	0.50	150
105	0.4	3	0	0.8	1	0.75	150
108	0.4	3	0	0.8	1	1.00	150

	Accuracy	Kappa	AccuracySD	KappaSD
1	0.8047438	0.6086276	0.007361343	0.014960645
4	0.8025259	0.6039578	0.003944092	0.007674354
7	0.8078163	0.6144657	0.005922455	0.012022202
10	0.8122514	0.6234408	0.012562724	0.025591520
13	0.8115693	0.6221969	0.008992123	0.018150997
16	0.8149847	0.6288576	0.007521131	0.015182657
55	0.8139598	0.6269251	0.010693929	0.021610136
58	0.8136175	0.6259536	0.009427523	0.018845925
61	0.8131063	0.6251494	0.006755793	0.013629500
64	0.8107173	0.6200285	0.015358670	0.031023842
67	0.8182259	0.6352089	0.009037010	0.018490039
70	0.8161810	0.6311720	0.009532753	0.019375511
19	0.8412672	0.6815355	0.009192630	0.018696471
22	0.8503130	0.6996989	0.012360128	0.024937870
25	0.8470719	0.6929576	0.005065493	0.010256479
28	0.8530459	0.7052882	0.003725782	0.007603659
31	0.8446818	0.6883166	0.006028018	0.012187358
34	0.8462185	0.6913306	0.012489091	0.025189862
73	0.8498037	0.6988434	0.003076943	0.006163964
76	0.8542391	0.7075779	0.012824553	0.025877618
79	0.8559469	0.7109905	0.007765175	0.015776054
82	0.8431451	0.6854275	0.013052023	0.026290715
85	0.8550915	0.7093306	0.013342412	0.026730471

88	0.8535587	0.7060663	0.008106272	0.016467986
37	0.8518518	0.7028674	0.011233483	0.022632412
40	0.8585081	0.7161959	0.010886437	0.022001167
43	0.8685763	0.7364000	0.012049493	0.024205124
46	0.8559487	0.7110475	0.004086332	0.008337927
49	0.8634570	0.7261911	0.009186900	0.018509765
52	0.8707952	0.7408009	0.010543198	0.021332601
91	0.8504860	0.7003611	0.009245676	0.018544612
94	0.8603851	0.7202582	0.003681792	0.007334104
97	0.8682372	0.7357850	0.007017761	0.014003734
100	0.8559481	0.7112156	0.005769243	0.011734928
103	0.8670418	0.7335327	0.004078825	0.008100407
106	0.8709662	0.7412810	0.006060476	0.012201733
2	0.8346122	0.6681501	0.012696212	0.025846648
5	0.8279540	0.6545234	0.010934856	0.022212248
8	0.8288084	0.6562177	0.007891043	0.016115243
11	0.8322214	0.6635437	0.009420375	0.018968770
14	0.8342708	0.6674510	0.006850892	0.014020365
17	0.8311976	0.6609936	0.010453142	0.021078632
56	0.8375154	0.6741808	0.008533858	0.017206426
59	0.8402442	0.6794605	0.008643424	0.017494362
62	0.8390489	0.6769215	0.008766174	0.017773247
65	0.8320517	0.6631636	0.012744704	0.025744801
68	0.8390499	0.6771284	0.005524696	0.011210447
71	0.8330759	0.6649139	0.007346259	0.014821196
20	0.8559482	0.7110653	0.006939474	0.013935181
23	0.8605554	0.7203387	0.008734105	0.017578029
26	0.8682355	0.7357088	0.008866176	0.017862897
29	0.8624330	0.7240724	0.011096010	0.022390765
32	0.8666995	0.7326461	0.006627005	0.013213732
35	0.8667014	0.7326777	0.011461137	0.023192360
74	0.8610685	0.7216587	0.006278640	0.012521302
77	0.8653333	0.7299620	0.011455304	0.023090799
80	0.8711381	0.7415428	0.003032802	0.006207282
83	0.8545832	0.7084121	0.011992885	0.024147901
86	0.8651650	0.7297581	0.008244738	0.016681014
89	0.8711364	0.7415472	0.009009298	0.018215306
38	0.8629460	0.7252135	0.014748084	0.029739695
41	0.8748937	0.7491945	0.004723231	0.009582440
44	0.8776244	0.7545639	0.010405291	0.020825622
47	0.8668709	0.7330103	0.008712979	0.017374368
50	0.8760856	0.7515416	0.009104776	0.018366173
53	0.8861572	0.7717747	0.005857941	0.011804499

92	0.8615808	0.7226576	0.003868251	0.007689618
95	0.8719918	0.7434980	0.009728582	0.019619412
98	0.8777938	0.7549620	0.007145812	0.014411194
101	0.8631154	0.7256835	0.009908245	0.019957099
104	0.8745520	0.7485085	0.007843481	0.015648156
107	0.8818894	0.7632144	0.006801697	0.013708588
3	0.8424626	0.6841650	0.012739871	0.025408772
6	0.8399030	0.6786783	0.008912048	0.018024129
9	0.8443387	0.6874234	0.012620615	0.025424297
12	0.8463884	0.6920352	0.008071562	0.016278425
15	0.8462179	0.6914537	0.007319313	0.014810236
18	0.8436574	0.6861131	0.010301883	0.020848461
57	0.8455365	0.6904241	0.001842126	0.003636039
60	0.8496304	0.6984746	0.010191870	0.020456730
63	0.8527025	0.7043111	0.011693029	0.023712927
66	0.8378554	0.6748201	0.006718856	0.013525101
69	0.8429766	0.6850720	0.007010357	0.014183755
72	0.8472408	0.6933994	0.011918303	0.024046795
21	0.8697722	0.7388287	0.006479472	0.013108648
24	0.8702828	0.7398278	0.013343730	0.026902471
27	0.8733555	0.7460723	0.007588401	0.015270722
30	0.8680661	0.7354473	0.017346859	0.034726238
33	0.8690887	0.7375426	0.005758355	0.011426010
36	0.8764301	0.7521909	0.009744010	0.019647673
75	0.8607275	0.7209422	0.008467548	0.016992045
78	0.8661877	0.7316773	0.010012722	0.020095204
81	0.8735276	0.7462781	0.006259483	0.012590372
84	0.8537288	0.7068031	0.013506410	0.027270527
87	0.8747220	0.7488666	0.007086234	0.014283107
90	0.8745514	0.7485383	0.010123109	0.020410356
39	0.8690880	0.7375466	0.012523489	0.025169516
42	0.8818919	0.7631778	0.005786714	0.011767961
45	0.8868408	0.7731182	0.008980096	0.017953474
48	0.8689197	0.7372109	0.007426847	0.015028363
51	0.8858150	0.7709185	0.009649431	0.019431463
54	0.8938384	0.7871721	0.006231258	0.012509273
93	0.8668718	0.7332141	0.004381709	0.008839012
96	0.8803540	0.7601977	0.007637828	0.015439191
99	0.8844502	0.7682290	0.009530645	0.019242764
102	0.8661874	0.7319081	0.008155323	0.016150528
105	0.8806966	0.7607895	0.003629713	0.007330050
108	0.8878638	0.7752352	0.006592183	0.013237102

```
   nrounds max_depth eta gamma colsample_bytree min_child_weight subsample
54     150         3 0.3     0              0.8                1         1

Model saved as 'Base_Models/XGBoostWithFeatureEngineering.rds'. To load it later, use:

loaded_model <- readRDS('Base_Models/XGBoostWithFeatureEngineering.rds')

BestTune hyperparameters saved as 'Base_Models_Data/XGBoost/BestHyperparams_XGBoostWithFeatu

Confusion Matrix and Statistics

          Reference
Prediction Drowsy Natural
   Drowsy     101      19
   Natural    294     327

               Accuracy : 0.5776
                 95% CI : (0.5411, 0.6135)
    No Information Rate : 0.5331
    P-Value [Acc > NIR] : 0.008237

                  Kappa : 0.1913

 Mcnemar's Test P-Value : < 2.2e-16

            Sensitivity : 0.2557
            Specificity : 0.9451
         Pos Pred Value : 0.8417
         Neg Pred Value : 0.5266
             Prevalence : 0.5331
         Detection Rate : 0.1363
   Detection Prevalence : 0.1619
      Balanced Accuracy : 0.6004

       'Positive' Class : Drowsy


Confusion matrix saved as 'Base_Models_Data/XGBoost/ConfusionMatrix_XGBoostWithFeatureEngine

Performance metrics saved as 'Base_Models_Data/XGBoost/Metrics_XGBoostWithFeatureEngineering
ROC Data for plot building saved as 'Base_Models_Data/XGBoost/ROCData_XGBoostWithFeatureEngi
```

```
ROC plot saved as 'Base_Models_Data/XGBoost/ROC_XGBoostWithFeatureEngineering.png'.
```

Random Forest, feature pre-processing

```
  mtry  Accuracy     Kappa AccuracySD    KappaSD
1    2 0.8992979 0.7976453 0.01383511 0.02794006
2   67 0.9095396 0.8187478 0.01511814 0.03044090
3 2303 0.8975914 0.7949730 0.01289225 0.02580866


  mtry
2   67


Model saved as 'Base_Models/RandomForestWithFeatureEngineering.rds'. To load it later, use:


loaded_model <- readRDS('Base_Models/RandomForestWithFeatureEngineering.rds')


BestTune hyperparameters saved as 'Base_Models_Data/Random_Forest/BestHyperparams_RandomFores


Confusion Matrix and Statistics

          Reference
Prediction Drowsy Natural
   Drowsy     345      98
   Natural     50     248

               Accuracy : 0.8003
                 95% CI : (0.7696, 0.8285)
    No Information Rate : 0.5331
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.5953

 Mcnemar's Test P-Value : 0.0001118

            Sensitivity : 0.8734
            Specificity : 0.7168
         Pos Pred Value : 0.7788
         Neg Pred Value : 0.8322
             Prevalence : 0.5331
         Detection Rate : 0.4656
```

```
   Detection Prevalence : 0.5978
      Balanced Accuracy : 0.7951

       'Positive' Class : Drowsy


Confusion matrix saved as 'Base_Models_Data/Random_Forest/ConfusionMatrix_RandomForestWithFea

Performance metrics saved as 'Base_Models_Data/Random_Forest/Metrics_RandomForestWithFeatureE
ROC data saved as 'Base_Models_Data/Random_Forest/ROCData_RandomForestWithFeatureEngineering

ROC plot saved as 'Base_Models_Data/Random_Forest/ROC_RandomForestWithFeatureEngineering.png
```

Data Metrics Combining

SVM classification report construction and best hyperparameter csv construction

XGBoost classification report construction and best hyperparameter csv construction

RandomForest classification report construction and best hyperparameter csv construction

Combine all to make the final base model report:

Combine Hyperparameters to a final report

AUCROC Plot Construction

SVM Plot

XGBoost Plot

Random Forest Plot

Stacking Classifier

Logistic Regression

logistic regression

```
Model saved as 'Stacked_Models/LogReg.rds'. To load it later, use:

loaded_model <- readRDS('Stacked_Models/LogReg.rds')
```

```
BestTune hyperparameters saved as 'Stacked_Models_Data/LogReg/BestHyperparams_LogReg.csv'.

[1] "Drowsy"  "Natural"

 Factor w/ 2 levels "Drowsy","Natural": 2 2 2 2 2 2 2 1 1 2 ...

[1] "Drowsy"  "Natural"

Confusion Matrix and Statistics

          Reference
Prediction Drowsy Natural
   Drowsy     349      10
   Natural     13     370

               Accuracy : 0.969
                 95% CI : (0.9539, 0.9803)
    No Information Rate : 0.5121
    P-Value [Acc > NIR] : <2e-16

                  Kappa : 0.938

 Mcnemar's Test P-Value : 0.6767

            Sensitivity : 0.9641
            Specificity : 0.9737
         Pos Pred Value : 0.9721
         Neg Pred Value : 0.9661
             Prevalence : 0.4879
         Detection Rate : 0.4704
   Detection Prevalence : 0.4838
      Balanced Accuracy : 0.9689

       'Positive' Class : Drowsy


Confusion matrix saved as 'Stacked_Models_Data/LogReg/ConfusionMatrix_LogReg.csv'.

Performance metrics saved as 'Stacked_Models_Data/LogReg/Metrics_LogReg.csv'.
ROC Data for plot building saved as 'Stacked_Models_Data/LogReg/ROCData_LogReg.csv'.
```

```
ROC plot saved as 'Base_Models_Data/SVM/ROC_SVMWithFeatureEngineering.png'.

Model saved as 'Stacked_Models/LinSVM.rds'. To load it later, use:

loaded_model <- readRDS('Stacked_Models/LinSVM.rds')

BestTune hyperparameters saved as 'Stacked_Models_Data/LinSVM/BestHyperparams_LinSVM.csv'.

[1] "Drowsy"  "Natural"

 Factor w/ 2 levels "Drowsy","Natural": 2 2 2 2 2 2 2 1 1 2 ...

[1] "Drowsy"  "Natural"

Confusion Matrix and Statistics

          Reference
Prediction Drowsy Natural
   Drowsy     344       8
   Natural     18     372

               Accuracy : 0.965
                 95% CI : (0.9491, 0.977)
    No Information Rate : 0.5121
    P-Value [Acc > NIR] : < 2e-16

                  Kappa : 0.9298

 Mcnemar's Test P-Value : 0.07756

            Sensitivity : 0.9503
            Specificity : 0.9789
         Pos Pred Value : 0.9773
         Neg Pred Value : 0.9538
             Prevalence : 0.4879
         Detection Rate : 0.4636
   Detection Prevalence : 0.4744
      Balanced Accuracy : 0.9646

       'Positive' Class : Drowsy
```

```
Confusion matrix saved as 'Stacked_Models_Data/LinSVM/ConfusionMatrix_LinSVM.csv'.

Performance metrics saved as 'Stacked_Models_Data/LinSVM/Metrics_LinSVM.csv'.
ROC Data for plot building saved as 'Stacked_Models_Data/LinSVM/ROCData_LinSVM.csv'.

ROC plot saved as 'Base_Models_Data/SVM/ROC_SVMWithFeatureEngineering.png'.
```

XGBoost Stacked

```
Model saved as 'Stacked_Models/XGBoost.rds'. To load it later, use:

loaded_model <- readRDS('Stacked_Models/XGBoost.rds')

BestTune hyperparameters saved as 'Stacked_Models_Data/XGBoost/BestHyperparams_XGBoost.csv'.

[1] "Drowsy"  "Natural"

 Factor w/ 2 levels "Drowsy","Natural": 2 2 2 2 2 2 2 1 1 2 ...

[1] "Drowsy"  "Natural"

Confusion Matrix and Statistics

          Reference
Prediction Drowsy Natural
   Drowsy     352      12
   Natural     10     368

               Accuracy : 0.9704
                 95% CI : (0.9555, 0.9813)
    No Information Rate : 0.5121
    P-Value [Acc > NIR] : <2e-16

                  Kappa : 0.9407

 Mcnemar's Test P-Value : 0.8312

            Sensitivity : 0.9724
            Specificity : 0.9684
```

```
         Pos Pred Value : 0.9670
         Neg Pred Value : 0.9735
             Prevalence : 0.4879
         Detection Rate : 0.4744
   Detection Prevalence : 0.4906
      Balanced Accuracy : 0.9704

       'Positive' Class : Drowsy


Confusion matrix saved as 'Stacked_Models_Data/XGBoost/ConfusionMatrix_XGBoost.csv'.

Performance metrics saved as 'Stacked_Models_Data/XGBoost/Metrics_XGBoost.csv'.
ROC Data for plot building saved as 'Stacked_Models_Data/XGBoost/ROCData_XGBoost.csv'.

ROC plot saved as 'Base_Models_Data/SVM/ROC_SVMWithFeatureEngineering.png'.
```

Random Forest Stacked

```
note: only 2 unique complexity parameters in default grid. Truncating the grid to 2 .

Model saved as 'Stacked_Models/RandomForest.rds'. To load it later, use:

loaded_model <- readRDS('Stacked_Models/RandomForest.rds')

BestTune hyperparameters saved as 'Stacked_Models_Data/RandomForest/BestHyperparams_RandomFo

[1] "Drowsy"  "Natural"

 Factor w/ 2 levels "Drowsy","Natural": 2 2 2 2 2 2 2 1 1 2 ...

[1] "Drowsy"  "Natural"
```

```
Confusion Matrix and Statistics

          Reference
Prediction Drowsy Natural
   Drowsy     353      15
   Natural      9     365

               Accuracy : 0.9677
                 95% CI : (0.9523, 0.9792)
    No Information Rate : 0.5121
    P-Value [Acc > NIR] : <2e-16

                  Kappa : 0.9353

 Mcnemar's Test P-Value : 0.3074

            Sensitivity : 0.9751
            Specificity : 0.9605
         Pos Pred Value : 0.9592
         Neg Pred Value : 0.9759
             Prevalence : 0.4879
         Detection Rate : 0.4757
   Detection Prevalence : 0.4960
      Balanced Accuracy : 0.9678

       'Positive' Class : Drowsy


Confusion matrix saved as 'Stacked_Models_Data/RandomForest/ConfusionMatrix_RandomForest.csv

Performance metrics saved as 'Stacked_Models_Data/RandomForest/Metrics_RandomForest.csv'.
ROC Data for plot building saved as 'Stacked_Models_Data/RandomForest/ROCData_RandomForest.cs

ROC plot saved as 'Base_Models_Data/SVM/ROC_SVMWithFeatureEngineering.png'.
```

Final Plot Constructions

References

[1] .

[2] Mario Sáenz Espinoza, Ana I Pereira, and José Gonçalves. "Optimization methods for hyper-redundant robots' inverse kinematics in biomedical applications". In: *AIP Conference Proceedings*. Vol. 1479. 2012, p. 818.

[3] Mario Sáenz Espinoza et al. "Inverse Kinematics of a 10 DOF modular hyper-redundant robot resorting to exhaustive and error-optimization methods: A comparative study". In: *Robotics Symposium and Latin American Robotics Symposium (SBR-LARS), 2012 Brazilian*. IEEE. 2012, pp. 125–130.

[4] Ming Tham. *Writing Research Theses or Dissertations*. University of Newcastle Upon Tyne. May 2001. URL: http://lorien.ncl.ac.uk/ming/dept/Tips/writing/thesis/thesis-intro.htm.

```
  )

# 9) Save the plot
ggsave(
  filename = "PlotsAndPictures/Combined_AUCROCPlots/StackedPlot.png",
  plot = p,
  width = 8,
  height = 6
)
```

Code Appendix